Liu Xingzhou

Email: 1246609140@qq.com

EDUCATION BACKGROUND

Nanjing University of Information Science and Technology (NUIST) ***Sept. 2023-Jun. 2027***
School of Mathematics & Statistics
Degree: Bachelor of Science in Mathematics and Applied Mathematics **GPA:** 86.29/100
English: IELTS 6.5
Computer: Python, MATLAB, LaTeX, MS Office
Certificate: National Computer Rank Examination (NCRE) Level II

RESEARCH & COMPETITION

Jiangsu Province College Students' Innovation and Entrepreneurship Training Program ***Apr. 2025-Apr. 2026***
***Leader*, Application and Exploration of Baltamatica Platform in Experimental Teaching Reform at Double First-Class Universities**

- Built a Baltamatica-based reusable experimental teaching system to replace foreign software in university mathematics courses and enhance students' scientific computing skills.
- Surveyed nearly 200 participants via questionnaires and interviews to identify teaching pain points, and compared Baltamatica with MATLAB across syntax, function libraries, visualization, and compatibility.
- Designed a five-step cycle teaching reform framework for platform feature mining, teaching pain point matching, script development, teaching process restructuring, and learning outcome evaluation.
- Created over 50 Baltamatica teaching scripts for 3D solid geometry visualization, linear algebra matrix transformations and eigenvalue graphics, and numerical analysis finite difference methods with error visualization.
- Piloted the teaching approach in three classroom sessions with about 30 students, collected over 200 survey responses, and found over 70% of respondents reported improved comprehension through dynamic visualization.

The 10th Dimension Cup National College Mathematical Contest in Modeling (National Excellence Award) ***May. 2025***
***Leader*, Marathon Economic Impact Analysis Based on Multi-Source Data Fusion and Intelligent Decision Models**

- Developed a data-driven decision-making framework for marathon events in major Chinese cities, and addressed event window selection, route planning, multi-objective optimization, and service design.
- Collected meteorological, demographic, traffic, and marathon registration data from 11 cities and Xi'an spatial datasets, and performed data cleaning, coordinate transformation, normalization, and spatial analysis with Python and ArcGIS.
- Built a TOPSIS model from meteorological indicators to determine optimal event months across cities, and applied K-means clustering with the elbow method to classify events into four tiers and generate frequency recommendations.
- Identified candidate start and end points via GIS analysis of subway proximity and accommodation capacity, and generated the 42km optimal route via DBSCAN clustering, improved ant colony algorithm, and BFS.
- Applied GIS-based tree shade quantification and cellular automata traffic simulation, and generated Pareto optimal routes via NSGA-II optimization to balance shade coverage, traffic disruptions, and slope compliance.

National College Students Statistical Modeling Competition ***Feb. 2025-Apr. 2025***
***Leader*, HPV Vaccination Rate Analysis and Intervention Strategy Optimization Based on Multi-Source Data**

- Investigated the gap between high HPV awareness and low vaccination rates in Nanjing, identified key factors in vaccination willingness, and developed an evaluation framework with actionable intervention strategies.
- Surveyed women aged 18–45 across 11 Nanjing districts, collected 525 valid responses (Cronbach's α=0.913, KMO=0.916), and built a multi-source dataset from published literature, health policy, and census data.
- Applied binary logistic regression to identify five significant factors in vaccination willingness and used FAHP to construct a three-tier evaluation system to rank factors by weight.
- Evaluated and weighted 25 indicators across knowledge, socioeconomic, and policy dimensions, and proposed targeted intervention strategies for vaccine supply, price subsidies, promotional channels, and market regulation.

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INTERNSHIP EXPERIENCE

Intern*, Nanjing Qunlu Cultural and Art Training Co., Ltd.** ***Mar 2026–Jun 2026

- Delivered one-on-one math and chemistry tutoring to 10 students (grades 10–12) on sets, functions, derivatives, geometry, sequences, and organic, inorganic, and analytical chemistry, and completed over 100 teaching hours.
- Tailored individualized lesson plans to each student's learning profile, communicated with parents to collect feedback, and refined instructional strategies and after-class support accordingly.

General Affairs Intern*, Inner Mongolia Autonomous Region Tobacco Company Xing'an League Branch** ***Jul. 2025-Aug. 2025

- Assisted in cigarette allocation and order verification for over 400 retailers in the jurisdiction, coordinated daily inquiries on supply cycles and product specifications, and ensured delivery accuracy.
- Streamlined departmental business processes and conducted classification, aggregation, and trend analysis of retailer order data with Excel to support marketing strategy adjustments.
- Tracked inventory reconciliation, retailer information entry, and feedback compilation, and ensured timely delivery of all milestones.

STUDENT WORK

Member*, Arts and Culture Department, Student Union, NUIST** ***Sep. 2023-Sep. 2024

- Organized more than 10 campus events, including the Mathematics Culture Festival, Campus Singing Competition, and Campus Music Festival.
- Served as a peer mentor in the "Huayang Classroom" program, supporting academic tutoring and experience-sharing activities for fellow students.

EXCHANGE EXPERIENCE

Summer Academic Program, The University of Tokyo ***Jul. 2024-Aug. 2024***

- Studied environmental governance, disaster prevention, aging society management, and educational systems in Japan through academic lectures, seminars, and field visits.
- Visited physics laboratories at The University of Tokyo, explored advanced research facilities and technologies in nuclear fusion and experimental physics, and gained exposure to interdisciplinary scientific research.

HONORS & AWARDS

Second-Class Scholarship of NUIST ***Dec. 2025 & Dec. 2024***

Outstanding Student of NUIST ***Dec. 2025 & Dec. 2024***

Third Prize, The Chinese Mathematics Competitions for College Students (Jiangsu Division) ***Nov. 2025***

Second Prize, NUIST Mathematical Modeling Competition ***Jun. 2025***